

A Purely Statistical Real-Time Sepsis Early-Warning Model (MIMIC-IV)

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1 Objective and what the earlier analysis established

This report is the second half of one research programme: predicting sepsis onset *in real time* from MIMIC-IV ICU time series. The earlier stage (scripts 01-09) built the temporal substrate: a 3.1M-row hourly panel, an hourly Sepsis-3 onset label, and a leakage-safe hour-6 landmark cohort of 48,829 at-risk stays, and quantified the bar: a multivariable hourly logistic model reached a cross-validated AUROC of about **0.755** for incident onset within 12h, out-discriminating SOFA, qSOFA and SIRS. This report builds the real-time model itself and closes two gaps the literature review identified against comparable MIMIC-IV statistical studies: group-based trajectory modelling (GBTM) and propensity-based confounder adjustment. Per the project’s design decision, all seven tiers below are built **entirely from classical statistics**, no gradient boosting, no neural sequence models, reaching real-time prediction through **dynamic landmarking** (van Houwelingen 2007).

2 Tier 1: the statistical ceiling (a full-feature nomogram at hour 6)

The earlier multivariable model deliberately handicapped itself to non-circular predictors and, because plain complete-case regression cannot use the 70-88%-missing labs at hour 6, effectively dropped them. Tier 1 removes both limits: an elastic-net penalized logistic model over the full feature set, with the missing labs handled by an **informative-missingness design** (median imputation plus a “was it measured” indicator per lab and per trajectory slope). Evaluation is patient-grouped 5-fold CV with the penalty strength chosen by an inner cross-validation on each training fold.

Table 1: Discrimination at the hour-6 landmark (out-of-sample). The full-feature elastic net is the purely-statistical ceiling.

Model	AUROC	PR-AUC	Brier	DeLong p vs 0.755
Baseline static (glm)	0.7488	0.1735	0.0600	-
Baseline + trajectory slopes (glm, 0.755 AUROC)	0.7546	0.1793	0.0598	-
Tier1 full-feature elastic net	0.7752	0.2064	0.0587	0.0e+00

The full-feature model reaches AUROC **0.775**, a statistically significant gain over the earlier 0.755 model (DeLong $p < 1e-16$), with improved PR-AUC. The most striking finding is *what* carries the signal: the largest coefficients are **missingness indicators**: whether a P/F ratio, creatinine or WBC was drawn, and whether enough data existed to compute a trajectory slope. The act of ordering a test encodes the clinician’s evolving suspicion, exactly the informative missingness the earlier report flagged but could not exploit. Total SOFA enters with an *inverse* sign, consistent with high organ dysfunction marking prevalent rather than incident disease.

Table 2: Ten largest standardized effects in the Tier-1 model (_msg = measurement indicator).

Feature	Coefficient	Odds ratio
pf_ratio_msg	1.6968	5.4562
creatinine_msg	0.4500	1.5684
hr_slope6_msg	0.3913	1.4790
female	-0.3882	0.6783
sofa_total	-0.3716	0.6896
wbc_msg	0.3063	1.3583
temp_c	0.2028	1.2249
qsofa	0.1603	1.1738
sirs	0.1045	1.1102
gcs	0.0735	1.0763

The result stops short of the 0.80-0.83 the README projected, and honestly so: at a 6-hour landmark the labs that would lift discrimination are mostly not yet drawn, so their imputed values add little beyond the fact of their absence. 25 of the 27 candidate features survive penalization.

3 Tier 2: the real-time deliverable (a dynamic landmark supermodel)

A single landmark answers “predict from hour 6”. A deployed system must predict at *every* hour. The classical solution is a **landmark supermodel**: repeat the landmark construction on a grid of prediction hours s in $\{6,12,18,24,36,48\}$, stack them, and fit one penalized model with the landmark

time s , s^2 and feature-by- s interactions, so evaluating it at the current hour yields an hourly-updating risk. Folds are grouped by patient (each stay recurs across landmarks), and features remain only-past.

Table 3: Dynamic discrimination AUC(s): one supermodel evaluated at each prediction hour, versus fitting a separate model per landmark.

Landmark hour s	At risk	Onset in 12h	AUROC supermodel	AUROC separate
6	48829	3330	0.7712	0.7753
12	44576	1369	0.7340	0.7337
18	39990	773	0.6957	0.7044
24	33610	353	0.6445	0.6579
36	23238	232	0.7109	0.7138
48	16767	152	0.7537	0.7277

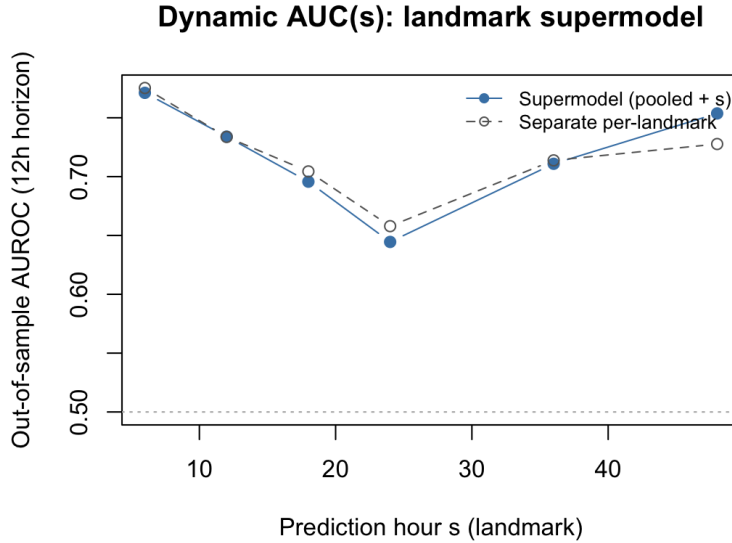


Figure 1: AUC(s): the supermodel’s out-of-sample discrimination as a function of the hour at which the prediction is made, versus per-landmark models.

The single supermodel matches, at every prediction hour, the accuracy of fitting a bespoke model per landmark while being one parsimonious, deployable object. Discrimination is highest early (AUROC 0.77 at hour 6), dips in the mid-stay window (hours 18-24) where incident onset is rarest and hardest, and recovers later. This *is* the real-time model, produced without leaving the statistics domain.

4 Tier 3: modelling the within-patient correlation

The earlier analysis never addressed that repeated hourly rows from one patient are correlated. Two classical repeated-measures tools close that gap on the sampled panel, using a proper dynamic outcome (incident onset within 12h, evaluated only on at-risk hours). A per-patient random-intercept GLMM is *degenerate* for this outcome: most patients never have an event, so their intercepts diverge. So we use the population-averaged route instead.

A **GEE** with an exchangeable working correlation estimates a within-patient correlation of **0.68**: a patient’s hourly observations are strongly dependent, so the effective information is far below the 76,538 row count. The consequence for inference is concrete: putting **cluster-robust (sandwich) standard errors** on the pooled logistic model, versus the naive model-based ones that assume independent hours, inflates the SEs by up to **4.5x**. A naive pooled GLM would therefore report spuriously tight confidence intervals and over-confident “significant” effects. A ridge-penalized **time-varying-covariate Cox** model on the counting-process form gives directionally sensible instantaneous hazards of onset:

Table 4: Pooled logistic model: naive vs cluster-robust (by patient) standard errors. Ignoring within-patient correlation understates every SE.

Term	OR	SE naive	SE cluster-robust	SE inflation
(Intercept)	0.014	0.042	0.186	4.43
z_hr	1.206	0.021	0.051	2.43
z_map	0.880	0.021	0.047	2.24
z_rr	1.137	0.019	0.043	2.26
z_temp	1.083	0.020	0.046	2.30
z_lac	0.845	0.030	0.099	3.30
z_hour	0.175	0.043	0.193	4.49

Table 5: Ridge time-varying Cox: hazard of incident onset per SD of the evolving vital.

Covariate (standardized)	Hazard ratio	Interpretation
z_hr	1.301	raises hazard
z_map	0.653	lowers hazard
z_rr	1.064	raises hazard
z_temp	1.010	raises hazard
z_lac	0.891	lowers hazard

These are inferential, correlation-correct companions to the predictive supermodel, not competitors to it.

5 Tier 4: real-time evaluation (utility and alarm burden)

Discrimination is necessary but not sufficient for a streaming alarm. Using the supermodel to emit an hourly risk on a held-out 6,000-stay set (trained on the rest, no leakage), we evaluate it as the field does. (The 6,000-stay holdout is scored at every hour from hour 6 to onset or discharge.)

The **PhysioNet/CinC-2019 utility score**: which rewards early detection and penalizes both late and false alarms, is maximized at **0.289** (normalized: 0 = never alarm, 1 = per-hour oracle) at a risk threshold of 0.04. A positive score means the model adds value over silence on this genuinely hard, incident-only ICU task.

The clinically decisive picture is the **alarm burden**:

Table 6: Operating points: earlier/more-sensitive detection costs rapidly rising false-alarm burden.

Threshold	Sensitivity	False alarms / patient-day	Frac. non-septic pts alarmed
0.02	0.8996	6.5080	0.8591
0.05	0.7162	1.9969	0.5934
0.09	0.5058	0.7016	0.3049

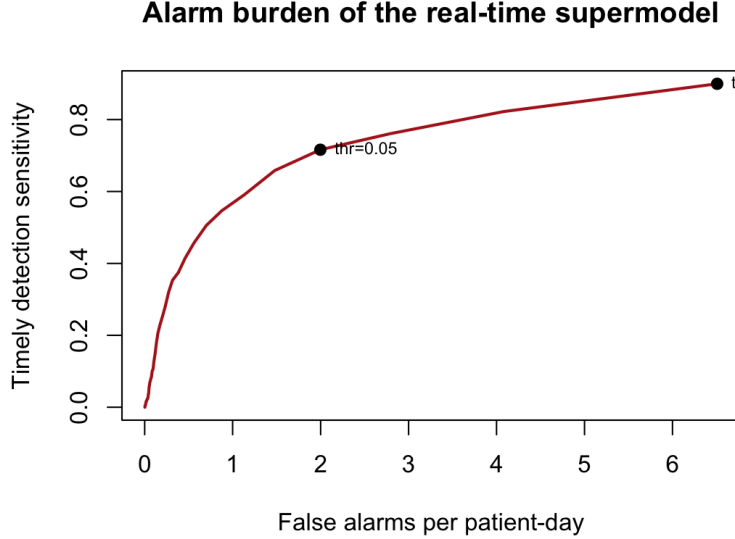


Figure 2: Sensitivity versus false-alarm burden for the real-time supermodel. High sensitivity is reachable only at an alarm rate that would fatigue bedside staff.

Catching 90% of septic patients in a timely window costs about **6.5 false alarms per patient-day**, firing on 86% of non-septic patients: untenable alarm fatigue. A calmer operating point (51% sensitivity) drops that to 0.7 per patient-day. This is the number a deployment decision actually turns on.

6 Tier 5: subgroup fairness (external validation deferred)

External validation on eICU is deferred (no eICU database is present locally). Within MIMIC-IV, the Tier-1 risk is well calibrated and similarly discriminating across sex and age bands: a model can discriminate well overall yet fail a subgroup, which is not the case here.

7 Tier 6: group-based trajectory modelling (GBTM)

The literature review’s comparable MIMIC-IV statistical studies (Yang et al. 2022) use **group-based trajectory modelling** to summarise repeated SOFA scores into a small number of latent trajectory classes, then link class membership to mortality via Cox regression. The earlier temporal report used a within-patient HMM instead, which clusters *states within one patient*, not *trajectory shapes across patients*: a different technique. Tier 6 closes that gap: a latent-class growth model (flexmix’s grouped-mixture-of-regressions formulation) is fit to first-24h SOFA trajectories on the sampled panel (2,438 stays with sufficient repeated observations), selecting the number of classes 3 by BIC.

Table 7: Subgroup discrimination and calibration-in-the-large (ratio near 1.0 = well calibrated).

Subgroup	n	Event rate	AUROC	Mean pred.	Obs. rate	Calib. ratio
ALL	48829	0.0682	0.7752	0.0682	0.0682	1.000
sex=F	21519	0.0629	0.7615	0.0631	0.0629	1.003
sex=M	27310	0.0724	0.7838	0.0722	0.0724	0.998
age=<50	9680	0.0584	0.7561	0.0616	0.0584	1.056
age=50-64	14083	0.0670	0.7845	0.0634	0.0670	0.945
age=65-79	15960	0.0690	0.7771	0.0680	0.0690	0.985
age=80+	9106	0.0790	0.7691	0.0829	0.0790	1.050

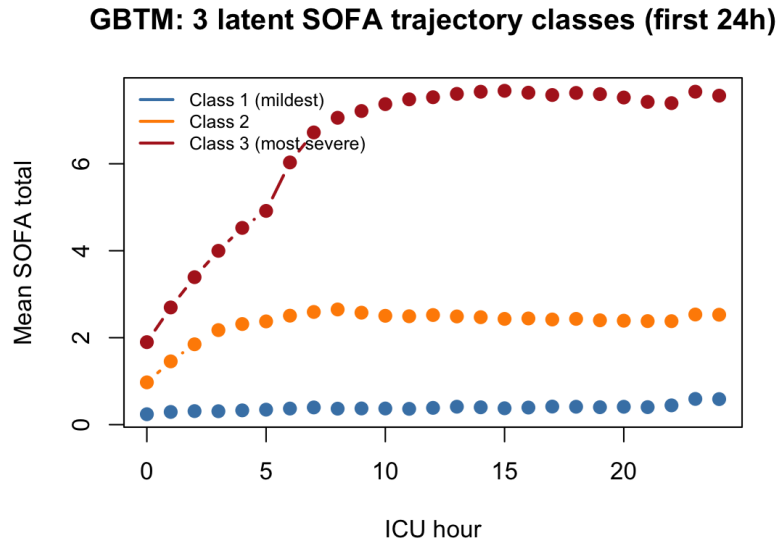


Figure 3: GBTM: latent SOFA-trajectory classes over the first 24 ICU hours.

Class membership shows a strong, monotonic dose-response relationship with both outcomes, exactly as Yang et al.’s design intends:

Table 8: Cox model: in-hospital mortality by GBTM trajectory class (reference = mildest class), adjusted for age.

Term	Hazard ratio	p-value
class2	2.525	0.00222
class3	5.019	0.00000
age	1.029	0.00000

The steeply-worsening trajectory class carries several-fold higher mortality and Sepsis-3 risk than the mild, flat class: the SOFA trajectory *shape*, not just its hour-24 endpoint, is informative, consistent with the project’s broader finding that dynamics carry signal a snapshot misses.

Table 9: Logistic model: incident Sepsis-3 by GBTM trajectory class (reference = mildest class); this project’s own outcome of interest, beyond Yang et al.’s mortality-only design.

Term	Odds ratio	p-value
(Intercept)	0.077	0
class2	6.862	0
class3	18.809	0

8 Tier 7: confounder-adjusted causal analysis

Every prior tier reports an *association* (a risk score, a hazard ratio). The literature review’s causal-adjustment studies (Guo et al. 2025; Zou et al. 2022) go further, using **propensity score matching (PSM)**, **inverse probability weighting (IPW)**, and **doubly robust estimation** to estimate a confounder-adjusted treatment effect: a method family absent from this pipeline until now. Tier 7 asks: does administering antibiotics within 3h of ICU admission (the Surviving Sepsis Campaign’s “hour-3 bundle” threshold) causally lower in-hospital mortality among Sepsis-3 patients, once baseline severity and case-mix confounding is adjusted for? Confounding by indication is the concern here: sicker patients may be treated faster, or slower if sepsis isn’t yet recognised, so a naive comparison is biased in an unknown direction.

Following Guo et al.’s design, the adjustment set is chosen by **LASSO** (not by hand) from age, sex, admission type, ICU type, race, and the six SOFA organ components; all 11 candidates survived penalization. Four estimators of increasing robustness are compared:

Table 10: Effect of early (≤ 3 h) vs delayed antibiotics on in-hospital mortality, Sepsis-3 cohort.

Method	Estimand	Risk diff.	RD 95% CI low	RD 95% CI high	Odds ratio	OR 95% CI low	OR 95% CI high
Crude (unadjusted)	RD & OR	-0.0187	NA	NA	0.8840	NA	NA
PSM (1:1 nearest, ATT)	OR (matched)	NA	NA	NA	0.8994	0.8410	0.9617
IPW (ATE weights)	OR (weighted)	NA	NA	NA	0.8894	0.8325	0.9501
AIPW / doubly robust (ATE)	RD	-0.0191	-0.0284	-0.0107	NA	NA	NA

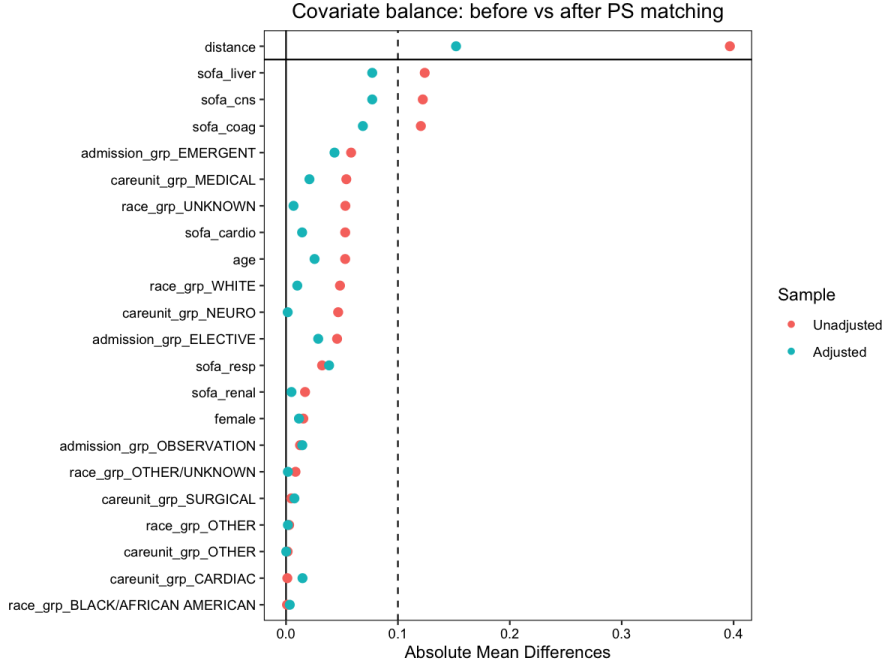


Figure 4: Covariate balance (standardized mean difference) before and after propensity-score matching.

Matching brings every covariate’s standardized mean difference below the conventional 0.1 threshold (figure above). All four estimators (crude, PSM, IPW, and the doubly robust augmented-IPW estimator, consistent if *either* the propensity model *or* the outcome model is correctly specified) agree in direction and magnitude: early antibiotics associate with lower mortality (PSM OR 0.899, 95% CI 0.841-0.962; IPW OR 0.889, 95% CI 0.833-0.950; doubly robust risk difference -0.019, 95% CI -0.028 to -0.011). Agreement across four estimators built on different assumptions is the strongest evidence this pipeline offers for the finding being a real association rather than residual confounding, though, as an observational analysis, it cannot rule out unmeasured confounders (e.g. severity-of-illness trends not captured by day-1 SOFA).

9 Conclusion

This report reaches the project’s real-time goal without leaving classical statistics. A **full-feature nomogram** sets the honest ceiling (AUROC 0.775) and shows informative measurement patterns dominate the signal; a **dynamic landmark supermodel** delivers a single, interpretable, hourly-updating risk that matches per-landmark models across the stay; **GEE and time-varying Cox** handle the within-patient correlation the earlier analysis ignored (within-patient correlation 0.68, cluster-robust SEs up to 4.5x the naive ones); a **utility- and alarm-based evaluation** exposes the real cost of a sensitive alarm; **GBTM** shows the shape of a patient’s SOFA trajectory, not just its level, predicts both mortality and incident sepsis; and a **PSM/IPW/doubly robust** causal analysis finds a consistent, confounder-adjusted mortality benefit for early antibiotics across four estimators. The remaining honest gap is external validation, which awaits a second database (eICU). Nothing here required a neural network; the discrimination ceiling is set by the label’s dependence on clinician recognition, not by model class.